

(+381) 654695389
Novi Sad, Serbia
tijanapetrovics03@gmail.com

Tijana Petrovic

Portfolio: [Tijana's-portfolio](#)
[github.com/tijanica](#)
[linkedin.com/in/tijana-petrovic-b8417a321](#)

Master's student at the Faculty of Technical Sciences, University of Novi Sad, with Bachelor's degree in Information Engineering, currently specializing in the Signal Processing module. My academic focus centers on data science, data processing, analytics, machine learning, and artificial intelligence. My approach to technical challenges relies on critical analysis and logical problem-solving.

SKILLS

Tools and Languages	Python, SQL, Java, MS Excel, PySpark, MongoDB, Neo4j, Docker, Git
Communication	English (Level C2), Serbian

TECHNICAL EXPERIENCE

Data Analyst Intern <i>Levi 9</i>	September 2024 <i>Novi Sad, Serbia</i>
---	--

- Used Pandas for local data manipulation, cleaning, and preprocessing.
- Employed PySpark to perform distributed data processing in cloud environments.
- Integrated AWS S3 for secure cloud data storage and retrieval.
- Developed interactive dashboards in Grafana for data visualization and monitoring.

Web Developer Intern <i>Synechron</i>	May 2025 — July 2025 <i>Novi Sad, Serbia</i>
---	--

- Developed a full-stack car insurance application, using React for the frontend and Spring Boot for the backend.
- Managed data persistence using MongoDB and MySQL, utilizing Git and GitLab for version control and team collaboration.

Substitute Informatics and Computing Teacher <i>Isidora Sekulic Grammar School</i>	March 2026 — May 2026 <i>Novi Sad, Serbia</i>
--	---

- Instructed students in core programming concepts, including Python, algorithms, and data structures.
- Covered foundational web development technologies (HTML, CSS, JavaScript) alongside SQL, data modeling, and MS Excel.

PROJECTS

Diabetes Prediction using ML <i>Python, Pandas, NumPy, Scikit-Learn, TensorFlow, Matplotlib, Seaborn</i>	GitHub Link
--	-----------------------------

- Developed a machine learning workflow to predict diabetes outcomes based on clinical datasets.
- Implemented and compared multiple classification algorithms, including Logistic Regression, Random Forests, Decision Trees, and Support Vector Machines (SVM).
- Performed data preprocessing, feature analysis, and exploratory data visualization using Matplotlib and Seaborn to identify key correlations.

Foodflow Delivery Application <i>Spring Boot, React, PostgreSQL, Figma, Jira</i>	GitHub Link
--	-----------------------------

- Designed and built a full-stack food delivery web application from the ground up.
- Modeled the relational database schema using ER diagrams and implemented data persistence with PostgreSQL.
- Created UI/UX wireframes in Figma and managed the project lifecycle using Jira to track sprints and progress.

CNN Neural Network Visualizer <i>Python, PyQt5, Matplotlib, NumPy, OpenCV, PyInstaller</i>	GitHub Link
--	-----------------------------

- Built an educational desktop application designed to visualize the architecture and operations of Convolutional Neural Networks (CNNs).
- Implemented 3D rendering of neural layers using Matplotlib and handled matrix calculations with NumPy.
- Integrated OpenCV for video input processing and packaged the tool into a standalone executable using PyInstaller.

Talking Faces in Serbian <i>Python, Wav2Vec 2.0, PyTorch, Transformers, GRUs, ARKit</i>	GitHub Link
---	-----------------------------

- Co-developed a multimodal, speech-driven facial animation system for Serbian, predicting 52 ARKit blendshape coefficients at 60 fps.
- Designed an architecture utilizing a fine-tuned Serbian Wav2Vec 2.0 feature extractor, triphone embeddings, a Transformer encoder, and a GRU decoder.
- Collected and processed a dataset of 808 speech utterances from five speakers, achieving a Pearson correlation of 0.866 between predicted and ground-truth shapes.

Microservice Travel Application

[GitHub Link](#)

C#, Go, React, Docker, Neo4j, MongoDB, MySQL

- Developed a Booking-like microservices platform designed for tourists and tour guides.
- Containerized distinct services using Docker to ensure consistent development environments and scalable deployments.
- Implemented a multi-database architecture, leveraging MySQL for transactional data, MongoDB for flexible documents, and Neo4j for relationship-based queries.

EDUCATION AND AWARDS

Bachelor with honours in Information Engineering, <i>Faculty of Technical sciencies, Novi Sad, Serbia</i>	October 2025
Dositej Obradovic Scholarship	2026
AI-Speak competition 2026 — 1st place award	AI-speak link